Manepalli Mohan Kumar

Curriculum Vitae — AI Engineer, Medical Imaging

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A results-driven AI Developer with 2 years of experience in **Deep Learning** and **Machine Learning**, including 1 year working on AI for **Medical Imaging**. I focused on **Deep Learning (DL)** for **Medical** and **Computer Vision (CV)** models, **Machine Learning (ML)**, **PyTorch** code building, and debugging errors during training and testing for tasks like **medical image segmentation**, classification, and **image translation**. My interests include **vascular disease screening**, **systemic disease detection**, and **fundus & OCT imaging**.

Education

Bachelor of Technology — **2022 — APRIL 2026**
Rajiv Gandhi University of Knowledge Technologies — *CGPA: 8.5*

Pre-University Course (Intermediate / MPC) — **2020 — 2022**
Rajiv Gandhi University of Knowledge Technologies — *CGPA: 9.56*

Secondary School Certificate (10th Class) — **2020**
BSE AP

- Secured the **State 64th Rank** in the 2020 Polytechnic Entrance Examination immediately following 10th grade.

Publications & Presentations

"Energy-Regularized Self-Supervised Denoising for Structure-Preserving OCT Imaging" — **MARCH 2026**
Optics Letters Journal — *Published*

- Addressed the **over-smoothing problem** in **S2SNet** for the **OCT dataset** by introducing energy regularization and validated improvements through a comprehensive comparison of **segmentation performance**.

"Data Driven Approach in Building Consumer-Friendly Business Models Using Machine Learning Techniques" — **DEC 2024**
***IEEE AICECS** Conference* — *Manipal Institute of Technology*

- Presented a research paper comparing three machine learning models on real-time dataset metrics.

Professional Experience

Technical Staff / TANUH (AICoE) IISc — **JUNE 2026 — PRESENT**
Vascular Disease Screening Vertical — *Bangalore, KA*

- Working in the **Vascular disease screening** vertical to detect **systemic diseases** from only **4 fundus images** per patient, leveraging **RETFound**, **MedGemma 1.5**, and **Qwen 3.8 27B** with **LoRA** fine-tuning.
- Built an **agent-based automated data quality check pipeline** to validate incoming clinical imaging data at scale.
- Developed and operated a **data collection and anonymisation website** (developer & IT owner) hosted on **GCP** across India for **real-time data collection from hospitals**.
- Administered the team's **compute nodes** and **Open OnDemand** software, managing access and workloads for researchers.

Engineering Intern / TANUH (AICoE) IISc — **NOV 2025 — MAY 2026**
Medical Assistant and Early Disease Detection — *Bangalore, KA*

- Working at **TANUH** (**AI Centre of Excellence** under the **Ministry of Education**) to develop **AI-driven diagnostics** and screening tools for early **breast cancer detection** and **risk prediction**.
- Developing "**Pseudohealth**" generative systems to synthesize **mammogram** images and **finetuned MedGemma 1.5 4B** to work with **mammogram** images, transforming clinical care through scalable AI solutions.

Developer / Video Masking & Annotation Application — **AUG 2025 — NOV 2025**
Full-Stack AI Deployment — *Bangalore, KA*

- Designed a medical video annotation tool integrating **SAM** for frame-by-frame analysis, deployed via a scalable serverless architecture on **Google Cloud Run**.

Research Intern / MIGLab [Link], CDS Department at IISc — **MARCH 2025 — OCT 2025**
***Medical Imaging** & Deep learning* — *Bangalore, KA*

- Executed distributed training via **torchrun** across **DGX Spark** clusters and optimized model inference through **Quantization** for real-time serving.
- Implemented **Flash Attention** and **MRS** pipelines for the **FeTS dataset** to train **3D UNet** models, optimizing data quality and training performance for large-scale vision tasks.
- Developed **OCT image denoising** algorithms and currently **developing model-based QSM reconstruction** algorithms, focusing on **preserving structural details** and high-precision accuracy.

Teaching Assistant / Aster Academy Cohort-5 — **2025**
AI in Healthcare / Medical Image Analysis — *Bangalore, KA*

- Supported instruction and technical experiments for a course on **AI in Healthcare** focused on **medical image analysis**.

Machine Learning Intern / Academor (Flutura) **JUNE 2023 — AUG 2023**
Remote

- Gained professional experience in applied machine learning, focusing on model internals, data curation, and version-controlled experimentation for industrial use cases.

Academic Projects

Deployment / Real-time Median-nerve Segmentation **JUNE 2025 — AUG 2025**
- Deployed **VisTr** on **Jetson Orin NX** using **NVIDIA SDK Manager**, while optimizing transformer performance with **Flash-Attn 2** and scaling training via **PyTorch DDP** on **dual RTX A6000 GPUs**.

Experimentation / Stable Diffusion 3.5 Medium & ComfyUI **2024**
- Executed inferencing of **Stable Diffusion 3.5** to generate high-quality images from text prompts utilizing the **ComfyUI** node-based interface on **Kaggle** infrastructure.

Developer / AgriBot (Collab with ICAR, Pune) **JAN 2024 — AUG 2024**
- Developed an end-to-end multimodal application for leaf disease detection and crop recommendation, integrating vision capabilities with **Arduino**, **ESP**, and **Raspberry Pi** for real-time inference.

Developer / Gemini AI-Powered Telegram Bot **DEC 2023**
- Built a multimodal agentic chatbot integrated with Google's Gemini API, utilizing **precise model configuration** for domain-specific user query answering.

Developer / Cloud-Based Automatic Object Finder **2023**
- Engineered a deep learning system for real-time multimodal object identification, utilizing cloud-streamed data and **YOLO** architectures.

Developer / LI-FI Prototype **2023**
- Developed a hardware-integrated prototype for high-speed wireless data transmission via light, demonstrating technical versatility in hardware-AI systems.

Technical Skills

Programming & Frameworks	Python (PyTorch, PyTorch Lightning, TensorFlow, Scikit-learn, MONAI), MATLAB, OpenCV, C/C++, HF Transformers, ComfyUI
LLMs & Generative AI	Vision Language Models (VLMs), Large Language Models (LLMs), Diffusion Models (DDPM), VAEs, GANs, Fine-tuning
DL Architectures	Transformer-based models (ViTs), Flash Attention, UNet, Multi-task & Multi-modal Learning
Imaging Systems & Infra.	Medical Imaging (MRI, CT, Endoscopy, Fundus, OCT), DICOM/NIfTI formats, Model Optimization, Edge/Mobile Architectures
CV & Processing	Image/Video Restoration, Enhancement, Denoising, Segmentation, Object Detection, OCR, Feature Extraction
Distributed Training & Infra.	DeepSpeed, torchrun, PyTorch, PyTorch Lightning DDP, GPU Acceleration, bfloat16, DGX Spark Clusters
Inference Optimization	Model Quantization (GGUF), Knowledge Distillation, vLLM, TensorRT
Algorithms & Concepts	Regression, Decision Trees, Random Forest, Gradient Boosting, Clustering, PCA, Neural Networks, CNNs
Experiment Tracking & DevTools	MLflow, TensorBoard, Python Logger, Git CLI
Communication	English, Telugu (Native), Hindi, Technical Documentation

Certifications

- Machine Learning for Engineering: NPTEL / IIT Madras [Link] 2025
- VLSI Design Flow: RTL to GDS: NPTEL / IIIT Delhi [Link] 2025
- Electronic Systems Design (PCB): NPTEL / IIT Madras [Link] 2025
- Python (Basic) Certified: HackerRank [Link] 2025
- Essential Mathematics for ML: NPTEL / IIT Roorkee [Link] 2024
- Deep Learning Specialist (PyTorch, TensorFlow, Keras): edX / IBM [Link] 2024
- Circuits & Electronics 3: edX / MITx [Link] 2024
- Geoprocessing using Python: IIRS / ISRO [Link] 2023
- Intro to Machine Learning: Kaggle [Link] 2023
- AI Appreciate & AI Aware (Combined): Intel [Link] 2023
- Pre-University Course (Intermediate / MPC): RGUKT [Link] 2022
- Secondary School Certificate (10th Class): BSE AP [Link] 2020

Activities

- Regularly analyze **Q1/Q2 journals** and state-of-the-art (**SOTA**) research to stay at the forefront of AI advancements.
- Participant: IEEE YESIST'12 Competition [Link] 2024
- National Service Scheme (NSS) 2021 — 2024